

The Vehicle Routing Problem with Time Windows

Chapter 5 — *Vehicle Routing: Problems, Methods, and Applications*

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Lecture Notes

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What is the VRPTW? I

The **Vehicle Routing Problem with Time Windows (VRPTW)** is one of the most studied and practically important variants of the classical Vehicle Routing Problem (VRP). In the standard VRP, a fleet of vehicles must deliver goods from a central depot to a set of geographically dispersed customers, minimising total travel distance or cost while respecting vehicle capacity limits. The VRPTW adds a crucial real-world constraint: each customer must be visited within a specified time interval called a *time window*.

Consider a concrete example. A grocery delivery service starts from a single warehouse (the depot) and must deliver to 100 households. Each household is only available to receive the delivery between, say, 8am and 10am (a hard constraint of two hours). The planning problem is to assign deliveries to vehicles, sequence them into routes, and ensure every household is served on time, while using as few vehicles as possible and keeping total driving distance low.

What is the VRPTW? II

VRPTW: Depot, Customers and Time Windows

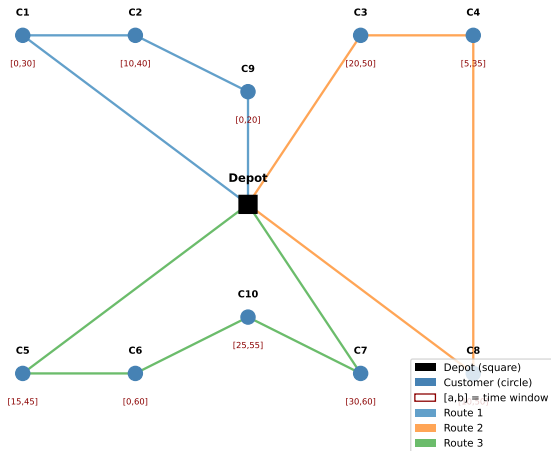


Figure: A VRPTW instance with 10 customers and 3 vehicle routes. Each customer (blue circle) has a time

Why the VRPTW Matters I

The VRPTW is the canonical model for a vast range of logistics operations that involve time-constrained delivery or service. Among the most common real-world applications are:

- **Urban delivery and courier services:** Parcels must reach customers during the home-delivery window the customer selected at checkout.
- **Home health care:** Nurses or therapists visit patients who are only available at specific times and require a given duration of care.
- **School bus routing:** Students must be picked up and dropped off within legally defined windows around school start times.
- **Industrial waste collection:** Bins overflow if not emptied within specified periods; spillage constitutes a constraint violation.
- **Airport ground operations:** Aircraft servicing (refuelling, catering, cleaning) must complete before departure, creating tight time windows.

Why the VRPTW Matters II

Key Insight

The time-window constraint transforms the VRPTW from a pure *distance-minimisation* problem into a *scheduling-and-routing* problem. Sequencing matters not just for distance but for feasibility: inserting a customer with an early deadline in the wrong position can make an otherwise short route infeasible.

Why the VRPTW Matters III

Computational complexity. The classical VRP without time windows is already NP-hard (it generalises the Travelling Salesman Problem). Adding time windows makes the problem even more constrained and, in the worst case, harder to solve. However, tight time windows also *prune* the search space significantly: many sequences are immediately infeasible, and this structured infeasibility is exploited by exact methods such as column generation.

Two objectives. In practice and in the literature, VRPTW solutions are evaluated on two criteria, listed in order of priority:

- ① **Primary:** Minimise the total number of vehicles used. Using fewer vehicles reduces fixed fleet costs.
- ② **Secondary:** Given the minimum number of vehicles, minimise the total travel distance or time.

This hierarchical objective is why VRPTW algorithms report both the vehicle count and the total distance of their best solutions.

Benchmark instances. Solomon (1987) introduced a widely used set of 56 benchmark instances with 100 customers each. These instances are classified by the spatial distribution of customers (random, clustered, or mixed) and by the width of the time windows (tight or wide), yielding six classes: C1, C2, R1, R2, RC1, RC2. Nearly all algorithmic comparisons in the VRPTW literature report results on these instances.

5.2 — Mathematical Formulations: Setup I

Before writing the mathematical model, we need to define all the variables and parameters carefully. This is important because the formulation guides both the exact solution methods and the design of heuristics.

5.2 — Mathematical Formulations: Setup II

Parameters and Sets

- n — number of customers (indexed 1 to n).
- $V = \{0, 1, \dots, n, n+1\}$ — node set; 0 and $n+1$ are the origin and destination depot copies.
- $V' = \{1, \dots, n\}$ — customer set (no depot).
- K — set of vehicles (indexed $1, \dots, m$).
- Q — vehicle capacity (maximum load, in units of demand).
- d_i — demand of customer i (units to deliver); $d_0 = d_{n+1} = 0$.
- $[a_i, b_i]$ — time window for customer i ; service must begin no earlier than a_i (alpha sub i) and no later than b_i (beta sub i).
- s_i — service time at customer i (duration of unloading).
- t_{ij} — travel time from node i to node j .
- c_{ij} — travel cost from node i to node j (often $= t_{ij}$ when minimising time, or Euclidean distance when minimising distance).

5.2 — Mathematical Formulations: Setup III

Decision Variables

- x_{ij}^k — binary variable equal to 1 if vehicle k travels directly from node i to node j , and 0 otherwise. (Pronounced “x sub ij superscript k .”)
- w_i^k — continuous variable representing the time at which vehicle k begins service at customer i . (“w sub i superscript k .”) If vehicle k arrives before the time window opens, it must wait; if it arrives after b_i , the solution is infeasible.

Note that x_{ij}^k captures the *routing* structure (which vehicle goes where, in what order), while w_i^k captures the *timing* (when each service starts). These two families of variables are tightly linked: the timing w_i^k is determined by the sequence encoded in x_{ij}^k together with travel and service times.

5.2 — The Three-Index VRPTW Formulation (Model I) I

The most transparent formulation uses three indices: one for the origin node i , one for the destination node j , and one for the vehicle k . This formulation is sometimes called the *vehicle-indexed arc formulation*.

Objective Function (5.1)

$$\min \sum_{k \in K} \sum_{(i,j) \in A} c_{ij} x_{ij}^k$$

This objective function says: minimise the total travel cost summed over all arcs (i,j) in the network and all vehicles k . Here c_{ij} is the cost of traversing arc (i,j) (its distance or travel time), and x_{ij}^k is the binary indicator that vehicle k uses that arc. Only arcs actually used by a vehicle contribute to the sum.

5.2 — The Three-Index VRPTW Formulation (Model I) II

How to Read This

Think of it this way: each vehicle “draws” a path on the map. The total cost is the sum of the distances of all arcs in all drawn paths. We want the drawing to be as short (cheap) as possible. Adding more vehicles reduces each route’s length but adds fixed fleet costs (if modelled); in the pure VRPTW, the primary objective is instead to minimise vehicle count, and distance is secondary.

5.2 — The Three-Index VRPTW Formulation (Model I) III

Constraints

(5.2) Each customer is visited exactly once:

$$\sum_{k \in K} \sum_{j \in V, j \neq i} x_{ij}^k = 1 \quad \forall i \in V'$$

For each customer i , exactly one vehicle must travel the outgoing arc from i . This ensures every customer is served by precisely one vehicle.

(5.3) Vehicles depart from the depot:

$$\sum_{j \in V'} x_{0j}^k = 1 \quad \forall k \in K$$

Each vehicle k leaves the origin depot exactly once. (Here we allow unused vehicles too—in practice we may relax this to ≤ 1 .)

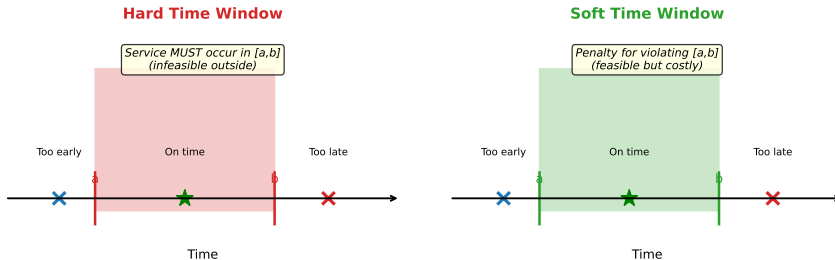
(5.4) Flow conservation at customer nodes:

$$\sum x_{ih}^k - \sum x_{hi}^k = 0 \quad \forall h \in V', k \in K$$

5.2 — Hard and Soft Time Windows I

The formulation above uses *hard time windows*: late arrival is simply infeasible (constraint 5.8 is inviolable). Many real-world problems instead use *soft time windows*, where violations are penalised but allowed.

Hard vs. Soft Time Windows



5.2 — Hard and Soft Time Windows II

Figure: Left: Hard time window. The vehicle must arrive in $[a, b]$; arriving outside this interval makes the solution infeasible. Right: Soft time window. Arriving outside $[a, b]$ is allowed but incurs a penalty proportional to the violation amount.

5.2 — Hard and Soft Time Windows III

Soft Time Window Penalty

For soft time windows, the objective function gains additional penalty terms. A common formulation adds, for each customer i and vehicle k :

$$\alpha_k^- \cdot \max(0, a_i - w_i^k) + \alpha_k^+ \cdot \max(0, w_i^k - b_i)$$

Here α^- (alpha, the Greek letter for “a”) is the penalty per unit of *early arrival* (the vehicle arrives before the window opens and must wait—a possible cost if waiting is undesirable), and α^+ is the penalty per unit of *late arrival* (the vehicle arrives after the window closes—a service failure with a financial penalty or customer dissatisfaction cost).

The symbol $\max(0, \cdot)$ means we only take a positive value if the violation is non-zero. When $w_i^k \geq a_i$ (arrival on time or late), the early penalty is zero; when $w_i^k \leq b_i$ (not late), the late penalty is zero.

5.2 — Hard and Soft Time Windows IV

Key Insight

Hard time windows are the standard assumption in the VRPTW literature, because they lead to cleaner formulations and allow meaningful comparison on Solomon benchmarks. Soft time windows are more common in practice, especially in customer-facing applications where a small lateness is tolerable at a known cost (e.g., a \$5 coupon for a late delivery). The choice between hard and soft profoundly affects which solution methods are applicable.

5.2 — The Two-Index Formulation (Model II) I

For many exact and heuristic methods, it is convenient to use a *vehicle-free* (two-index) formulation in which the vehicle index k is dropped. This formulation is valid when all vehicles are homogeneous (identical capacity and cost structure) and the objective is separable over arcs.

Objective (5.10)

$$\min \sum_{(i,j) \in A} c_{ij} x_{ij}$$

where $x_{ij} \in \{0, 1\}$ indicates whether arc (i, j) is used by *any* vehicle (we no longer track which specific vehicle uses it).

5.2 — The Two-Index Formulation (Model II) II

Key Constraints

(5.11) Each customer visited once:

$$\sum_{j \in V \setminus \{i\}} x_{ij} = 1 \quad \forall i \in V'$$

(5.12) Flow conservation:

$$\sum_{i \in V} x_{ij} = \sum_{i \in V} x_{ji} \quad \forall j \in V'$$

(5.13) Timing consistency (linearised):

$$w_j \geq w_i + s_i + t_{ij} - M(1 - x_{ij}) \quad \forall (i, j) \in A$$

Here M (a large number, called the “big- M ”) is a sufficiently large constant that when $x_{ij} = 0$, the constraint is trivially satisfied. When $x_{ij} = 1$, it enforces the timing logic: $w_j \geq w_i + s_i + t_{ij}$.

5.2 — The Two-Index Formulation (Model II) III

Two-Index vs. Three-Index: Trade-offs

The three-index formulation (Model I) has $O(n^2m)$ binary variables and $O(n^2m)$ constraints. For large instances with many vehicles, this is huge. The two-index formulation has $O(n^2)$ binary variables but loses the ability to enforce vehicle-specific constraints (e.g., different capacities per vehicle). In practice, the *set-partitioning* formulation described next is often preferred for exact methods because its LP relaxation is much tighter.

Number of Vehicles

The minimum number of vehicles is sometimes enforced as a lower bound constraint. If we want at most m vehicles:

$$\sum_{j \in V'} x_{0j} \leq m$$

This is the constraint that “at most m routes depart from the depot.” In hierarchical VRPTW, vehicle minimisation is the *primary* objective, so one first solves for the minimum number of vehicles and then minimises distance with that fixed fleet size.

5.2 — Set Partitioning Formulation (Model III) I

The *set partitioning* (or *path-based*) formulation is the foundation of the most powerful exact algorithms. Instead of modelling individual arcs, it operates over the set of all feasible routes.

5.2 — Set Partitioning Formulation (Model III) II

Set-Partitioning Model

Let $\mathcal{R}$ be the set of all feasible routes (paths from depot 0 to depot $n + 1$ that respect capacity and time windows). For each route $r \in \mathcal{R}$:

- c_r — total travel cost of route r .
- $a_{ir} \in \{0, 1\}$ — equals 1 if customer i appears on route r , and 0 otherwise.
- λ_r (lambda sub r , the Greek letter lambda) — binary decision variable equal to 1 if route r is selected in the solution.

Objective:

$$\min \sum_{r \in \mathcal{R}} c_r \lambda_r$$

Coverage constraints (each customer served exactly once):

$$\sum_{r \in \mathcal{R}} a_{ir} \lambda_r = 1 \quad \forall i \in V'$$

Integrality:

$$\lambda_r \in \{0, 1\} \quad \forall r \in \mathcal{R}$$

5.3 — Exact Methods: Overview I

Because the VRPTW is NP-hard, exact methods can only solve instances of limited size (up to around 100 customers in practice, with great difficulty). Nevertheless, exact methods are important because they (a) prove optimal solutions for benchmark instances, establishing reference points; (b) provide lower bounds for heuristic quality; and (c) generate structural insights that inform heuristic design.

The main exact approaches for the VRPTW are:

- ➊ **Branch-and-Bound with arc-based LP relaxation.** An early approach that solves the LP relaxation at each node of a branch-and-bound tree. The LP relaxation is weak, so many nodes are needed.
- ➋ **Column generation** (solving the LP relaxation of the set-partitioning formulation). This is the foundation of modern exact methods. Instead of enumerating all routes, it dynamically generates only the routes with potential to improve the solution.
- ➌ **Branch-and-Price (B&P).** Combines branch-and-bound with column generation at each node. This is the *state-of-the-art* exact approach and can solve instances with up to about 100 customers.
- ➍ **Branch-and-Price-and-Cut (B&P&C).** Adds cutting planes to tighten the LP relaxation, further reducing the tree size.

5.3.1 — Column Generation I

Column generation is the key technique that makes it tractable to solve the LP relaxation of the set-partitioning formulation. The core idea is: instead of including all exponentially many routes in the LP from the start, begin with a small *initial set* of routes (columns) and iteratively *add* only routes that can improve the current LP solution.

5.3.1 — Column Generation II

The Column Generation Loop

- 1 **Solve the Restricted Master Problem (RMP):** Solve the LP relaxation with the current set of columns (routes). Obtain dual variables π_i (π sub i , Greek letter π) for each customer i 's coverage constraint.
- 2 **Solve the Pricing Sub-problem:** Find a feasible route r with *negative reduced cost*:

$$\bar{c}_r = c_r - \sum_{i \in r} \pi_i < 0$$

Here $\bar{c}_r$ (c -bar sub r) is the reduced cost. The term c_r is the route's travel cost; the term $\sum_{i \in r} \pi_i$ is the “value” of the customers on that route according to the current dual prices π_i .

- 3 **Add column:** If such a route is found, add it to the RMP and return to step 1.
- 4 **Optimality certificate:** If no route with negative reduced cost exists, the current RMP solution is the optimal LP solution.

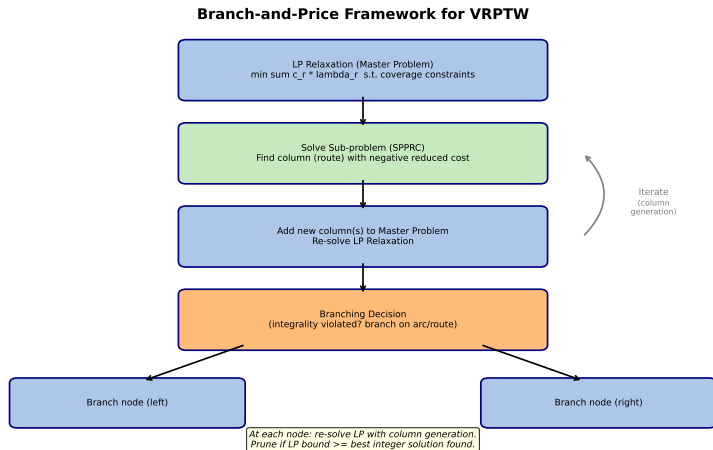
5.3.1 — Column Generation III

How to Read the Reduced Cost Formula

The reduced cost $\bar{c}_r = c_r - \sum_{i \in r} \pi_i$ measures whether adding route r to the solution can reduce the total objective. The dual variable π_i represents the “shadow price” of serving customer i : how much the objective would change if customer i ’s coverage requirement were relaxed. A route with $\bar{c}_r < 0$ means its cost is *less* than the combined value of serving its customers—so including it is beneficial.

Intuitively: if the LP is currently paying a lot to serve customer i (high π_i), then any route that cheaply includes i has a negative reduced cost and should be considered.

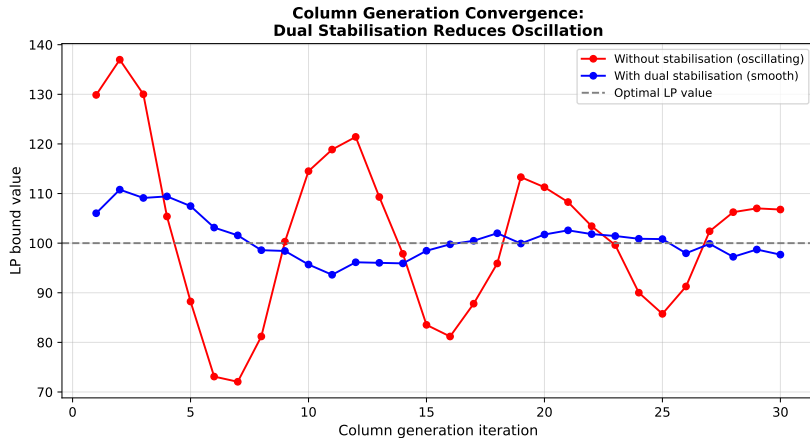
5.3.1 — Column Generation IV



5.3.1 — Column Generation V

Figure: The Branch-and-Price framework. Column generation solves the LP relaxation at each branch-and-bound node. When the LP is optimal, a branching decision is made on a fractional variable.

5.3.1 — Column Generation VI



5.3.1 — Column Generation VII

Figure: Column generation convergence. Without dual stabilisation (red), the LP bound oscillates because dual variables swing wildly between iterations. With stabilisation techniques such as bundle methods or proximal-point terms (blue), convergence is smoother and faster.

Dual stabilisation. A known practical difficulty with column generation is that dual variables can oscillate severely between iterations, especially early in the process. This slows convergence dramatically. Stabilisation methods—such as adding a *proximal term* to the RMP objective or restricting dual variables to a box around their current values—damp these oscillations and accelerate convergence. Dual stabilisation was introduced for the VRPTW by Rousseau, Gendreau, and Feillet (2007).

5.3.1 — The Pricing Sub-problem: SPPRC I

The pricing sub-problem—finding a feasible route with minimum reduced cost—is itself a non-trivial combinatorial problem. For the VRPTW, it is formulated as the **Shortest Path Problem with Resource Constraints (SPPRC)**.

SPPRC: What Problem Does It Solve?

We seek the least-cost path from the origin depot (node 0) to the destination depot (node $n + 1$) in the network G , where the path must satisfy two *resource* constraints at every intermediate node:

- 1 **Time resource:** The earliest feasible service start time at each node must fall within the node's time window $[a_i, b_i]$.
- 2 **Load resource:** The cumulative demand picked up along the path must not exceed the vehicle capacity Q .

The “cost” of an arc (i, j) in the pricing sub-problem is its *reduced travel cost* $c_{ij} - \pi_j$ (since the reduced cost of including node j on the route reduces the route's reduced cost by π_j).

5.3.1 — The Pricing Sub-problem: SPPRC II

Resource Extension Functions

The SPPRC is solved via *dynamic programming over labels*. A label at node j is a tuple (T_j, Q_j) where:

- T_j is the earliest feasible time at which the vehicle can start serving customer j (respecting earlier time windows on the path so far).
- Q_j is the total demand of customers visited so far on the path.

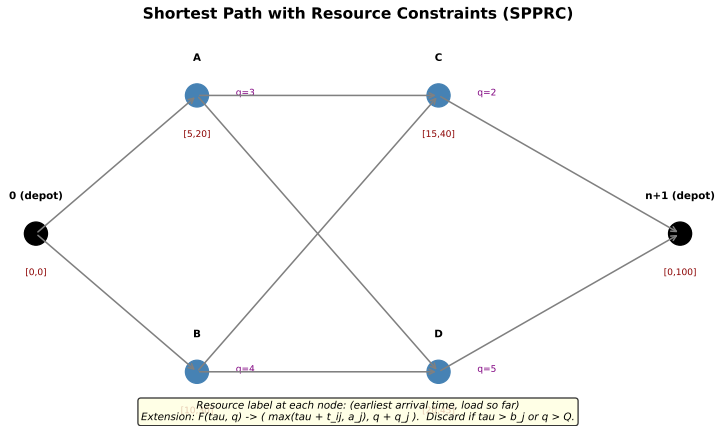
When extending a path from node i to node j , the label is updated by the *resource extension functions* (REFs):

$$T_j = \max(T_i + s_i + t_{ij}, a_j) \quad Q_j = Q_i + d_j$$

The first equation says: the earliest arrival time at j is the current time (T_i) plus service time at i (s_i) plus travel time from i to j (t_{ij}), but if we arrive before the window opens, we must wait until a_j . The second equation simply adds the demand of customer j .

A label is *infeasible* if $T_j > b_j$ (late arrival) or $Q_j > Q$ (capacity exceeded); such labels are discarded.

5.3.1 — The Pricing Sub-problem: SPPRC III



5.3.1 — The Pricing Sub-problem: SPPRC IV

Figure: SPPRC on a small network. The depot (node 0) and destination depot (node $n + 1$) are shown in black; customers A, B, C, D are shown in blue. Each node has a time window (red) and demand (purple). Dynamic programming propagates resource labels along arcs, pruning infeasible extensions.

5.3.1 — The Pricing Sub-problem: SPPRC V

Label Dominance (Pruning)

To avoid an exponential blowup in labels, we apply *dominance*: label $L_1 = (T_1, Q_1)$ *dominates* label $L_2 = (T_2, Q_2)$ at the same node if

$$T_1 \leq T_2 \quad \text{and} \quad Q_1 \leq Q_2$$

This means L_1 is at least as good as L_2 on *both* resources. Any extension of L_2 could also be achieved (and at least as well) by extending L_1 . Therefore L_2 can be discarded without losing optimality.

With dominance pruning, the labelling algorithm runs in pseudo-polynomial time. For each node j , we maintain only the set of *non-dominated* labels (the Pareto front of labels in (T, Q) space).

5.3.1 — The Pricing Sub-problem: SPPRC VI

Complexity Note

The SPPRC with time windows and capacity is NP-hard in general (it is equivalent to solving a constrained shortest path, which is NP-hard when there are two or more resources). However, the tight time windows in VRPTW instances severely restrict the number of non-dominated labels at each node, making the labelling algorithm tractable in practice. Proposition 1 (Desaulniers et al.) states that in bi-directional labelling, the two-label representation suffices.

5.3.1 — Labelling Algorithm for SPPRC I

The labelling algorithm is a generalisation of Dijkstra's algorithm that tracks resource consumption along each path.

5.3.1 — Labelling Algorithm for SPPRC II

Algorithm 1 Forward Labelling Algorithm for SPPRC

- 1: Initialise: at node 0 (depot), create label $L_0 = (T = 0, Q = 0, \text{cost} = 0)$.
- 2: Mark all nodes unprocessed. Add L_0 to the label set of node 0.
- 3: **while** there exists an unprocessed label **do**
- 4: Select an unprocessed label $L = (T_i, Q_i)$ at node i with minimum cost.
- 5: Mark L as processed.
- 6: **for** each arc $(i, j) \in A$ **do**
- 7: Compute extended label: $T_j \leftarrow \max(T_i + s_i + t_{ij}, a_j)$
- 8: $Q_j \leftarrow Q_i + d_j$
- 9: **if** $T_j \leq b_j$ **and** $Q_j \leq Q$ **then**
- 10: Create new label $L' = (T_j, Q_j, \text{cost}_L + \bar{c}_{ij})$
- 11: **if** L' is not dominated by any label at j **then**
- 12: Add L' to label set of j ; remove dominated labels.
- 13: **end if**
- 14: **end if**
- 15: **end for**
- 16: **end while**
- 17: **return** label at node $n + 1$ with minimum cost (= minimum reduced cost route).

5.3.2 — Branch-and-Price I

Column generation solves the LP relaxation of the set-partitioning formulation. To obtain an integer (combinatorial) solution, we embed column generation inside a **branch-and-bound** framework. The combination is called **Branch-and-Price (B&P)**.

Why Not Just Round the LP Solution?

The LP relaxation of the set-partitioning model allows λ_r to be fractional (e.g., $\lambda_r = 0.5$, meaning half a route). Rounding these values up or down typically produces infeasible or heavily suboptimal solutions. The branch-and-bound tree systematically explores all integer solutions implied by the LP relaxation, pruning branches that cannot improve on the best integer solution found so far.

5.3.2 — Branch-and-Price II

Branching Strategies for VRPTW

Standard branching on LP variables ($\lambda_r = 0$ or 1) destroys the pricing sub-problem structure. Instead, two branching strategies are commonly used:

- ① **Branch on arc flow:** Force arc (i, j) to be used (or not) by at least one vehicle. Formally:
$$\sum_{r: (i, j) \in r} \lambda_r \geq 1 \text{ or } = 0.$$
 This modifies the SPPRC by forbidding or requiring specific arcs.
- ② **Branch on the number of vehicles serving a cluster:** Group customers into a cluster and branch on whether the number of vehicles serving that cluster is $\leq k$ or $\geq k + 1$.

The most popular strategy (Desrochers, Desrosiers, and Solomon 1992) branches on arc flow, which can be imposed directly in the SPPRC by removing or requiring arcs.

5.3.2 — Branch-and-Price III

Valid Inequalities to Strengthen the Formulation

Various cutting planes can be added to tighten the LP relaxation at each B&P node:

- **Infeasible path inequalities:** Forbid sequences of arcs that are jointly infeasible due to time windows. For a sequence $S = (i_1, i_2, i_3)$ where visiting all three in order is time-infeasible, add:

$$\sum_{r: S \subseteq r} \lambda_r = 0$$

- **k -path cuts (subset-row inequalities):** For any subset $S \subseteq V'$ of customers, the minimum number of routes needed to cover S provides a lower bound on the number of routes intersecting S :

$$\sum_{r \in \mathcal{R}} \lfloor |r \cap S|/2 \rfloor \cdot \lambda_r \leq |K|$$

(where $|K|$ is the fleet size). These cuts are called *subset-row inequalities* (SRIs) or k -path cuts and significantly tighten the LP.

- **Capacity cuts:** Standard Gomory-Hu capacity inequalities from the VRP.

5.3.3 — Reduced Cost Variable Fixing I

An important technique to speed up Branch-and-Price is *reduced cost variable fixing* (also called *reduced cost pricing*). The idea is to eliminate arcs or routes from consideration *before* solving the full pricing sub-problem, by exploiting the current LP dual solution.

Principle

Suppose the current LP optimal value is z_{LP}^* and the best known integer solution has value $\bar{z}$. For any arc (i, j) , define its reduced cost as:

$$\bar{c}_{ij} = c_{ij} - \pi_i - \mu_j$$

where π_i (pi sub i) and μ_j (mu sub j , Greek letter mu) are dual variables. If $z_{LP}^* + \bar{c}_{ij} > \bar{z}$, then using arc (i, j) in any route cannot improve the integer solution. Such arcs can be *fixed to zero* (removed from the network), reducing the SPPRC complexity.

In practice, reduced cost fixing can eliminate 50–90% of arcs in tight instances, dramatically accelerating the pricing sub-problem.

5.4 — Heuristics: Overview and Motivation I

Exact methods for the VRPTW become computationally intractable for large instances (more than 100–150 customers). Real-world logistics instances regularly involve hundreds to thousands of customers. **Heuristics** sacrifice the guarantee of finding the optimal solution in exchange for finding high-quality solutions quickly.

The heuristic landscape for VRPTW divides naturally into three levels:

- 1 **Construction heuristics:** Build a feasible solution from scratch. They do not improve an existing solution; they produce a starting point. Examples: Solomon's insertion algorithms I1 and I2 (1987).
- 2 **Local search and improvement heuristics:** Start from a feasible solution and iteratively improve it by making small modifications (*moves*). Examples: 2-opt, Or-opt, cross-exchange. They are fast but can get trapped in local optima.
- 3 **Metaheuristics:** Higher-level frameworks that guide local search to escape local optima. Examples: Tabu Search, Simulated Annealing, Genetic Algorithms, Large Neighbourhood Search. These are the methods that achieve best-known solutions on Solomon benchmarks.

5.4.1 — Solomon's Insertion Heuristics (I1, I2) I

Solomon (1987) proposed the first systematic construction heuristics for the VRPTW. The most important is the **I1 algorithm**: a sequential insertion procedure that builds one route at a time and inserts customers based on a scoring function that balances distance and time window proximity.

Intuition. Starting from an empty route (just the depot), we repeatedly ask: which unrouted customer, if inserted at the best feasible position in the current route, causes the least additional cost? We insert that customer, update the route's timing, and repeat until no more customers can be feasibly inserted. Then we open a new route for the remaining customers.

5.4.1 — Solomon's Insertion Heuristics (I1, I2) II

I1 Algorithm: Criterion Functions

For inserting customer u between currently consecutive customers i and j on the route:

Cost criterion $c_1(i, u, j)$: the extra distance caused by the insertion:

$$c_1(i, u, j) = \alpha_1 \cdot (d_{iu} + d_{uj} - d_{ij}) + \alpha_2 \cdot b_u^* - b_j^*$$

Here d_{ij} (d sub ij) is the distance between i and j ; b_u^* (b -star sub u) is the new service start time at u if inserted; b_j^* (b -star sub j) is the resulting new service start time at j after the insertion. The parameters α_1 (alpha sub 1) and α_2 (alpha sub 2) weight the distance increase against the time-push effect (how much the insertion delays later customers). Typically $\alpha_1 + \alpha_2 = 1$.

5.4.1 — Solomon's Insertion Heuristics (I1, I2) III

I1 Algorithm: Selection Criterion

Selection criterion $c_2(u)$: which unrouted customer u to insert. The best position for u is the one that minimises c_1 ; call this minimum cost $c_1^*(u)$. Among all unrouted customers, we select the one with maximum

$$c_2(u) = \lambda \cdot d(\text{depot}, u) - c_1^*(u)$$

Here λ (lambda) is a parameter. The term $d(\text{depot}, u)$ is the distance from the depot to customer u ; subtracting the best insertion cost $c_1^*(u)$ biases selection toward customers that are *far from the depot but cheap to insert*—these are the customers who would be most expensive to handle in a future route if not inserted now.

5.4.1 — Solomon's Insertion Heuristics (I1, I2) IV

I1 Algorithm: Pseudocode

```
1:  $\mathcal{U} \leftarrow \{1, \dots, n\}$  (all customers unrouted)
2: while  $\mathcal{U} \neq \emptyset$  do
3:   Select seed customer:  $u_0 \leftarrow \arg \max_{u \in \mathcal{U}} d(\text{depot}, u)$  (or earliest deadline)
4:   Create new route:  $R \leftarrow (\text{depot} \rightarrow u_0 \rightarrow \text{depot})$ ;  $\mathcal{U} \leftarrow \mathcal{U} \setminus \{u_0\}$ 
5:   repeat
6:     for each  $u \in \mathcal{U}$  do
7:       Find best feasible insertion position  $(i^*, j^*)$  in  $R$  minimising  $c_1(i, u, j)$ 
8:     end for
9:     if any feasible insertion exists then
10:      Insert  $u^* = \arg \max c_2(u)$  at position  $(i^*, j^*)$ ;  $\mathcal{U} \leftarrow \mathcal{U} \setminus \{u^*\}$ 
11:    end if
12:   until no feasible insertion exists or  $\mathcal{U} = \emptyset$ 
13: end while
14: return set of routes
```

5.4.1 — Solomon's Insertion Heuristics (I1, I2) V

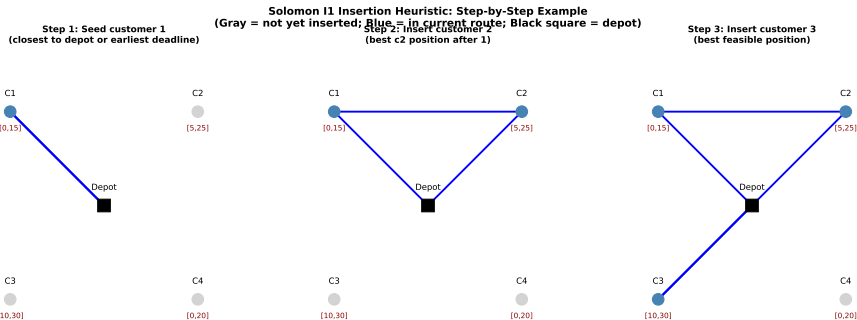


Figure: Three steps of the I1 insertion heuristic on a 4-customer example. Grey customers have not yet been inserted; blue customers are on the current route. At each step, the customer with the best c_2 score is inserted at its cheapest feasible position.

5.4.1 — Solomon's Insertion Heuristics (I1, I2) VI

Worked Numerical Example: I1 with 4 Customers

Consider 4 customers at positions $C1=(2,8)$, $C2=(8,8)$, $C3=(2,2)$, $C4=(8,2)$, depot at $(5,5)$. Time windows: $C1=[0,15]$, $C2=[5,25]$, $C3=[10,30]$, $C4=[0,20]$. Demands: all = 2; capacity $Q = 6$. Service time $s_i = 1$ for all. Parameters $\alpha_1 = 0.5$, $\alpha_2 = 0.5$, $\lambda = 1$.

Step 1: Seed. Compute $d(\text{depot}, C_i)$: $d_1 = \sqrt{9+9} \approx 4.24$, $d_2 \approx 4.24$, $d_3 \approx 4.24$, $d_4 \approx 4.24$ (symmetric; tie broken by earliest deadline). Seed: $C1$ (earliest $b_1 = 15$). Route: $\text{dep} \rightarrow C1 \rightarrow \text{dep}$.

Step 2: Insert C2. Only position is between $C1$ and depot.

$c_1 = 0.5(d_{C1,C2} + d_{C2,\text{dep}} - d_{C1,\text{dep}}) + 0.5(b_{C2}^* - b_{\text{dep}}^*)$. $d_{C1,C2} = 6$, $d_{C2,\text{dep}} = 4.24$, $d_{C1,\text{dep}} = 4.24$. Extra distance = $6 + 4.24 - 4.24 = 6$. Check: $w_{C1} = 2$, $w_{C2} = 2 + 1 + 6 = 9 \in [5, 25]$. Feasible.

Step 3: Continue inserting $C4$ and $C3$ into the route or a new route, always checking capacity ≤ 6 and time windows. When inserting $C3$ into the current route would violate the capacity (load would become 6), we check if the time windows still allow it.

5.4.1 — Neighbourhood Search Methods I

A *neighbourhood* of a solution s is a set $N(s)$ of solutions reachable from s by applying a single *move operation*. Local search starts from a feasible solution and repeatedly moves to a better neighbour.

Intra-route neighbourhoods (moves within a single route):

- **2-opt**: Remove two arcs (i, i') and (j, j') from the route and reconnect the four endpoints by adding arcs (i, j) and (i', j') , thereby reversing the segment between i' and j . Accept the move if the new route is feasible and shorter.
- **Or-opt** (also called **1-opt** or **relocation**): Move a single customer (or a segment of 2 or 3 consecutive customers) to a different position in the same route.

5.4.1 — Neighbourhood Search Methods II

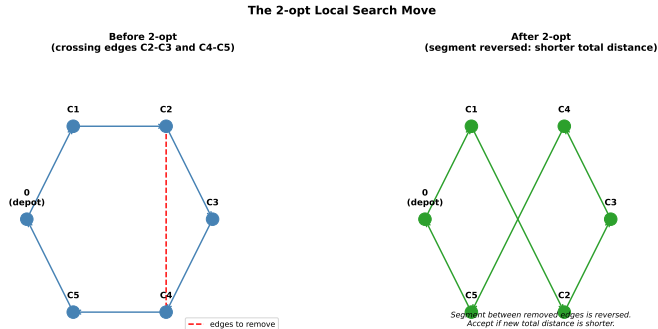


Figure: The 2-opt move. Two crossing arcs (dashed red) are replaced by two non-crossing arcs. The segment between the two removed arcs is reversed. This always reduces total distance if the arcs were crossing.

5.4.1 — Neighbourhood Search Methods III

Inter-route neighbourhoods (moves across two or more routes):

- **Or-opt (inter-route):** Move a customer from one route to a different route. This is the key move for reducing vehicle count.
- **Cross-exchange:** Exchange a segment of consecutive customers between two routes. This can create better temporal alignment when routes have complementary time windows.
- **2-opt* (two-opt-star):** A modified 2-opt that swaps the *tails* of two routes: all customers from position p onward in route 1 are transferred to route 2, and vice versa. Unlike standard 2-opt, 2-opt* operates across routes.
- **Path relocation:** Move a connected segment of k customers (a *path*) from one route to another.

5.4.1 — Neighbourhood Search Methods IV

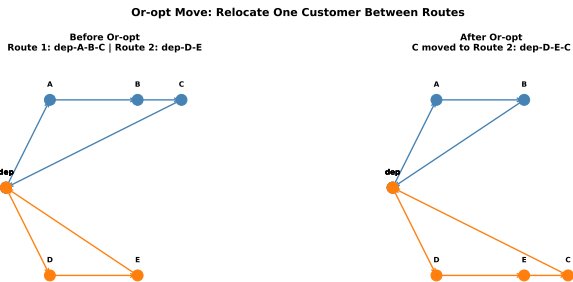


Figure: Or-opt move: customer C is removed from Route 1 and inserted into Route 2. This reduces Route 1's load and may allow a vehicle to be eliminated.

5.4.1 — Neighbourhood Search Methods V

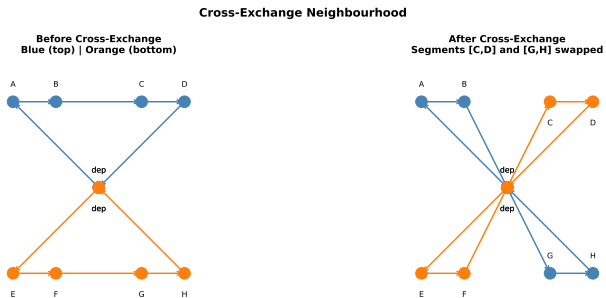


Figure: Cross-exchange: segment $[C, D]$ from the blue route is swapped with segment $[G, H]$ from the orange route. Both routes must remain feasible after the swap (time windows and capacity must still be satisfied).

5.4.1 — Neighbourhood Search Methods VI

Key Insight: Time Window Feasibility After Moves

When performing any neighbourhood move in the VRPTW, all affected routes must be checked for time-window feasibility after the move. This is more expensive than in the pure VRP. Efficient feasibility checking using *time-window propagation* (updating arrival times incrementally from the point of modification) is crucial for practical local search performance. For a route of n customers, a naive re-check costs $O(n)$; with proper data structures (e.g., storing earliest and latest feasible times), many moves can be evaluated in $O(1)$.

5.4.2 — Single Trajectory Search: Local Search and ILS I

The simplest metaheuristic framework is **single trajectory search**: at each step, apply a neighbourhood move and decide whether to accept it. The decision rule distinguishes different methods.

5.4.2 — Single Trajectory Search: Local Search and ILS II

Algorithm 5.1: Large Neighbourhood Search (LNS)

```
1:  $s \leftarrow$  initial feasible solution (e.g., from I1 heuristic)
2:  $s^* \leftarrow s$  (best solution found so far)
3: repeat
4:    $s' \leftarrow \text{Destroy}(s)$ 
5:    $s' \leftarrow \text{Repair}(s')$ 
6:   if  $f(s') < f(s)$  then
7:      $s \leftarrow s'$ 
8:   end if
9:   if  $f(s') < f(s^*)$  then
10:     $s^* \leftarrow s'$ 
11:  end if
12: until stopping criterion (e.g., iteration limit or time limit)
13: return  $s^*$ 
```

▷ Remove a subset of customers from s
▷ Re-insert removed customers into s'

5.4.2 — Single Trajectory Search: Local Search and ILS III

LNS intuition. The classical local search neighbourhoods (2-opt, Or-opt) are *small*: they change only a tiny portion of the solution. Large Neighbourhood Search (Shaw 1998) argues that it is more productive to destroy a large portion of the current solution (removing 10–30 customers simultaneously) and reconstruct it. The large destruction step “unlocks” the solution from local minima that small moves cannot escape.

5.4.2 — Single Trajectory Search: Local Search and ILS IV

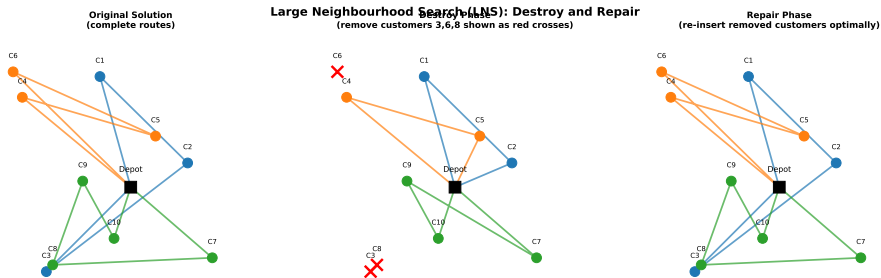


Figure: LNS applied to a 10-customer instance. Left: initial feasible solution. Middle: three customers (C3, C6, C8) are removed (red crosses) in the destroy phase. Right: the removed customers are re-inserted using a greedy repair heuristic. The repair may produce a different (and hopefully better) assignment.

5.4.2 — Single Trajectory Search: Local Search and ILS V

Adaptive LNS (ALNS)

Ropke and Pisinger (2006) extended LNS to **Adaptive LNS (ALNS)**, which maintains *multiple destroy and repair operators* and adaptively adjusts the probability of using each operator based on its historical performance.

Specifically, each operator ϕ (phi, Greek letter) has a weight w_ϕ . After each iteration, the weight is updated as:

$$w_\phi \leftarrow (1 - r) \cdot w_\phi + r \cdot \sigma_\phi$$

where r (rho, a reaction parameter) controls how quickly the algorithm adapts, and σ_ϕ (sigma sub phi) is a score based on how well operator ϕ performed (e.g., $\sigma = 3$ if the new solution is a new global best; $\sigma = 2$ if it improves the current solution; $\sigma = 1$ if it is accepted; $\sigma = 0$ otherwise). The selection probability of operator ϕ at each step is proportional to w_ϕ . This is analogous to a multi-armed bandit problem.

5.4.2 — Single Trajectory Search: Local Search and ILS VI

Why “Adaptive” Helps

Different destroy/repair operators work better on different instance types. For clustered instances (C1, C2), removing a geographic cluster of customers is effective; for random instances (R1, R2), a random removal works well. ALNS automatically discovers which operators work and allocates more probability to them. This explains why ALNS consistently outperforms fixed-operator LNS.

5.4.2 — Iterated Local Search (ILS) I

Iterated Local Search (ILS) is another single-trajectory framework that applies a *perturbation* to escape local optima.

ILS for VRPTW

```
1:  $s_0 \leftarrow$  initial solution (e.g., from IL);  $s \leftarrow \text{LocalSearch}(s_0)$ 
2:  $s^* \leftarrow s$ 
3: repeat
4:    $s' \leftarrow \text{Perturb}(s)$                                 ▷ Apply a random large move (e.g., random 3-opt)
5:    $s'' \leftarrow \text{LocalSearch}(s')$                           ▷ Re-apply local search from perturbed solution
6:    $s \leftarrow \text{Accept}(s, s'')$                                ▷ Accept if better, or by some probabilistic criterion
7:   if  $f(s'') < f(s^*)$  then
8:      $s^* \leftarrow s''$ 
9:   end if
10: until stopping criterion
11: return  $s^*$ 
```

5.4.2 — Iterated Local Search (ILS) II

Key idea. ILS alternates between deep local search (finding a local optimum) and perturbation (jumping to a different basin of attraction). The perturbation must be strong enough to escape the current basin but not so strong that it destroys all structure. For VRPTW, effective perturbations include random Or-opt moves or random segment cross-exchanges.

5.4.3 — Population-Based Search: Evolutionary Algorithms I

Evolutionary algorithms (EAs) maintain a *population* of solutions (rather than a single current solution) and improve them by applying *selection*, *recombination* (crossover), and *mutation* (local improvement).

General EA Framework for VRPTW

- 1: Generate initial population P of N feasible solutions (using I1 or random)
- 2: **repeat**
- 3: **Selection:** Choose two parent solutions $s_a, s_b \in P$ (e.g., tournament selection)
- 4: **Recombination:** Apply crossover operator to s_a, s_b to produce offspring s'
- 5: **Improvement:** Apply local search to s' to find local optimum
- 6: **Replacement:** Insert s' into P ; remove worst individual (or apply diversity criterion)
- 7: **until** stopping criterion
- 8: **return** best solution in P

5.4.3 — Population-Based Search: Evolutionary Algorithms II

Crossover Operators for VRPTW

Recombination in VRPTW must produce feasible children. Two effective operators are:

- **Order Crossover (OX):** Copy a segment of the customer sequence from parent s_a into the child; fill the remaining positions with customers from parent s_b in their original order. This preserves the relative ordering of customers from both parents.
- **Edge Assembly Crossover (EAX):** Identify arcs that appear in parent s_a but not s_b (and vice versa) and systematically reassemble routes using those arcs. EAX tends to preserve more route structure and is particularly effective for VRPTW.
- **Split + recombination:** Apply Prins' *split* algorithm: represent each solution as a giant tour (all customers in a single sequence), apply crossover on the giant tour, then use dynamic programming (split) to partition it optimally into routes respecting capacity and time windows.

5.4.3 — Population-Based Search: Evolutionary Algorithms III

EAX on a Small VRPTW Instance

Parent 1: Route A = [1, 2, 3], Route B = [4, 5]. Parent 2: Route C = [1, 4, 2], Route D = [3, 5].

EAX identifies “AB-cycles”: arcs in Parent 1’s routes that form cycles when composed with Parent 2’s arcs. By selecting which cycles to inherit from each parent, offspring with novel route combinations are created while preserving partial sub-sequences from both parents.

5.4.3 — Metaheuristic Overview: Tabu Search I

Tabu Search (TS), introduced by Glover (1986), is one of the most successful metaheuristics for the VRPTW. The analogy is this: a problem-solver exploring a landscape is allowed to “revisit” previously seen solutions, but maintains a *tabu list* of recently visited solutions or moves that are temporarily forbidden. This prevents cycling and forces exploration of new areas.

5.4.3 — Metaheuristic Overview: Tabu Search II

Tabu Search for VRPTW

- 1: $s \leftarrow$ initial feasible solution; $s^* \leftarrow s$
- 2: $\mathcal{T} \leftarrow \emptyset$ (tabu list, length $|\mathcal{T}| = \ell$)
- 3: **repeat**
- 4: Generate candidate moves $N(s)$ (e.g., all Or-opt, 2-opt* moves)
- 5: $s' \leftarrow \arg \min_{s'' \in N(s) \setminus \mathcal{T}} f(s'')$ ▷ Best non-tabu move
- 6: **if** $f(s') < f(s^*)$ (aspiration criterion) **then**
- 7: Accept s' even if tabu; $s^* \leftarrow s'$
- 8: **end if**
- 9: $s \leftarrow s'$
- 10: Add move generating s' to $\mathcal{T}$; remove oldest if $|\mathcal{T}| > \ell$
- 11: **until** stopping criterion
- 12: **return** s^*

5.4.3 — Metaheuristic Overview: Tabu Search III

Key parameters of Tabu Search:

- **Tabu list length ℓ :** Short lists allow rapid escape from local optima but risk cycling; long lists enforce more diversification but may prevent beneficial moves. Dynamic tabu list length (adapted based on solution quality) is common.
- **Aspiration criterion:** A tabu move is permitted if it produces a new global best. This is nearly universal in TS implementations.
- **Diversification:** After a long period without improvement, reset the solution or force exploration of under-visited areas.
- **Infeasibility handling:** Many VRPTW TS implementations allow temporary infeasibility (e.g., slight capacity violations) with a penalty, then repair the solution. This enlarges the neighbourhood and prevents getting trapped by feasibility constraints.

5.4.3 — Metaheuristic Overview: Tabu Search IV

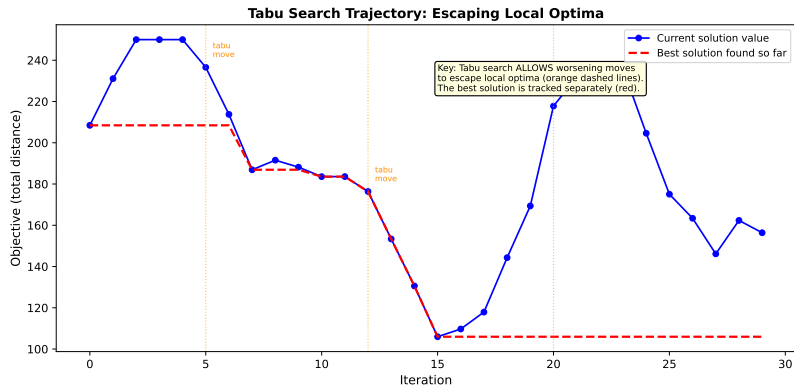


Figure: Tabu search trajectory. The current solution value (blue) is allowed to worsen at tabu moves (orange vertical lines), enabling escape from local minima. The best solution found (red dashed line) monotonically improves.

5.4.3 — Metaheuristic Overview: Tabu Search V

Allowing Infeasibility in Tabu Search

A key innovation in VRPTW tabu search (e.g., Taillard et al. 1997; Cordeau, Laporte, Mercier 2001) is to allow *temporarily infeasible* solutions during search. Instead of hard constraints on capacity and time windows, violations are penalised in the objective function. The algorithm oscillates between feasible and infeasible regions, achieving much richer exploration. The penalty weights are tuned dynamically: if the search has been infeasible for too long, increase penalties; if it has been overly constrained, reduce them.

5.4.3 — Simulated Annealing (SA) and Guided Local Search (GLS) I

Simulated Annealing (SA). The analogy comes from metallurgy: when metal is slowly cooled (*annealed*), atoms settle into low-energy crystal structures. Similarly, SA starts with a high “temperature” T (theta-like parameter) that allows accepting worse solutions with high probability, then gradually reduces T , progressively narrowing the search to improve solutions.

SA Acceptance Rule

A candidate solution s' (generated by a random neighbourhood move from s) is accepted if:

$$f(s') < f(s) \quad (\text{improvement, always accepted}) \quad \text{or} \quad \text{Uniform}[0, 1] < \exp\left(\frac{f(s) - f(s')}{T}\right)$$

Here T is the current temperature. When T is large, $\exp(\cdot)$ is close to 1 and even large worsening moves are accepted. As $T \rightarrow 0$, the acceptance probability of worse moves approaches 0 and SA reduces to greedy descent.

5.4.3 — Simulated Annealing (SA) and Guided Local Search (GLS) II

Cooling schedule: A common geometric cooling schedule is $T \leftarrow \beta \cdot T$ after each iteration (where β (beta) is slightly less than 1, e.g., $\beta = 0.9999$). The initial temperature T_0 and final temperature T_f must be tuned carefully.

5.4.3 — Simulated Annealing (SA) and Guided Local Search (GLS) III

Guided Local Search (GLS). Proposed by Voudouris and Tsang (1999), GLS augments the objective function with penalty terms for features of solutions that have been visited often without improvement.

GLS Augmented Objective

$$h(s) = f(s) + \lambda \sum_{i=1}^m p_i \cdot I_i(s)$$

Here m is the number of solution features (e.g., individual arcs), p_i (p sub i) is the penalty for feature i (incremented each time feature i appears in a local optimum), $I_i(s)$ (indicator sub i) equals 1 if feature i is present in solution s , and λ (lambda) is the penalty weight. After reaching each local optimum of h , the penalty of the feature with highest utility _{i} = $I_i(s) \cdot c_i / (1 + p_i)$ is incremented by 1.

The key insight is that GLS *remembers* which arcs or sub-routes appear repeatedly in local optima and penalises them, forcing the search to explore qualitatively different solutions. Prins and Bouchenoua (2004) applied GLS successfully to the VRPTW.

5.4.3 — Metaheuristic Comparison Table I

The following table summarises the key features of the most important metaheuristics for the VRPTW, based on Table 5.2 from the book (comparison of published methods).

Method	Ref.	2-opt	Or-opt*	Cross	LNS	Vehicle min.
SA	BVH04	×	×	×		×
TS	MB05			×		×
GLS	PR07				×	×
ILS	HY08	×	×	×		
LNS/CE	IINSUY08				×	
ALNS	RT09	×		3	×	×
EA+LNS	PDR09			3	×	
ILS+LNS	NBD10	×	×	1	×	×
Hybrid	VCGP13		×	2	×	×

5.4.3 — Metaheuristic Comparison Table II

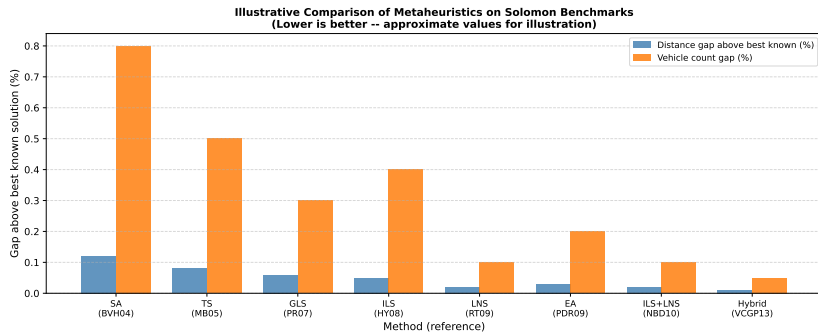


Figure: Illustrative comparison of metaheuristic quality on Solomon benchmarks. Lower bars indicate smaller gap above the best known solution. Hybrid methods combining multiple neighbourhoods and LNS (right side) consistently achieve the smallest gaps.

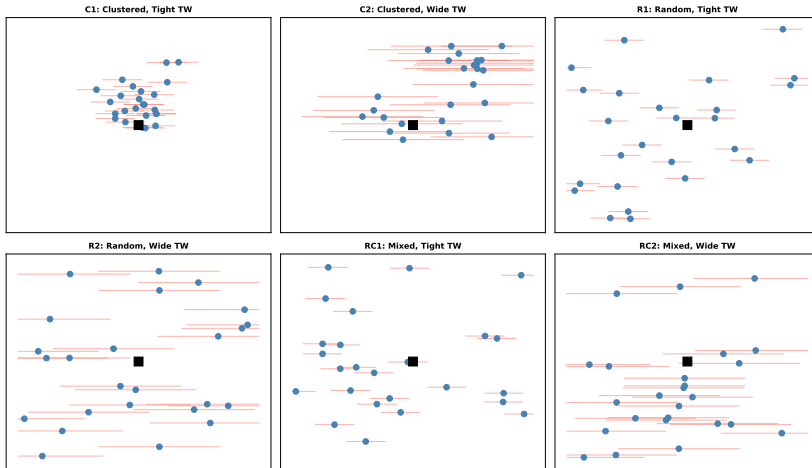
5.5 — Summary Computational Results for Exact Methods I

Table 5.3 in the book reports results of the best exact Branch-and-Price algorithms on Solomon instances. The instances are grouped into two classes:

- **Class 1** (C1, R1, RC1): tight time windows, many vehicles.
- **Class 2** (C2, R2, RC2): wide time windows, few vehicles.

5.5 — Summary Computational Results for Exact Methods II

Solomon (1987) Benchmark Instance Classes
(Black square = depot; blue dots = customers; red bars = time windows)



5.5 — Summary Computational Results for Exact Methods III

Figure: Solomon benchmark instance classes. Clustered instances (C1, C2) have customers grouped around centres. Random instances (R1, R2) distribute customers uniformly. Mixed instances (RC1, RC2) combine both. Red bars illustrate time window widths: tight in class 1, wide in class 2.

5.5 — Summary Computational Results for Exact Methods IV

Key findings from computational experiments (Table 5.3):

- **Class 1 instances** (tight windows): the Branch-and-Price algorithm of Desrochers, Desrosiers, Solomon (DDS92) solved all 25 instances optimally. Modern B&P+C methods require only a small fraction of the computation time compared to early methods.
- **Class 2 instances** (wide windows): the SPPRC sub-problem is harder (fewer dominated labels) and the tree is larger. However, with dual stabilisation and subset-row inequalities, all 25 Solomon instances have been solved optimally.
- The Chabrier (2006) B&P algorithm and the Righini-Salani (2006) bidirectional labelling are decisive advances. Matching lower bounds with LNS upper bounds confirms optimality.
- For instances with 200–400 customers (Gehring and Homberger benchmark), exact methods remain too slow; best results come from metaheuristics.

5.5 — Summary Computational Results for Exact Methods V

Benchmark Normalisation

Results are typically reported as CNV (total number of vehicles across all instances), CD (total travel distance), and CT (CPU time in seconds). Comparing algorithms is done on the *same* computing environment; historical comparisons must account for Moore's Law (computers double in speed roughly every 2 years).

Python: Solomon I1 Construction Heuristic

Listing 1: Solomon I1 insertion heuristic for VRPTW

```
import math

def euclidean(p1, p2):
    return math.hypot(p1[0]-p2[0], p1[1]-p2[1])

def i1_vrptw(depot, customers, Q, alpha1=0.5, alpha2=0.5, lam=1.0):
    """
    Solomon I1 heuristic for VRPTW.
    customers: list of dicts with keys x,y,demand,tw_open,tw_close,service_time
    depot: dict with keys x,y,tw_open,tw_close
    Q: vehicle capacity
    Returns list of routes (each route = list of customer indices)
    """
    n = len(customers)
    unrouted = set(range(n))
    routes = []

    while unrouted:
        # Select seed: farthest from depot (or earliest deadline)
        seed = max(unrouted,
                   key=lambda i: euclidean(
                       (depot['x'], depot['y']),
                       (customers[i]['x'], customers[i]['y'])))
        route = [seed]
        unrouted.remove(seed)
```

Python: 2-opt Improvement for VRPTW Route

Listing 2: 2-opt local search for a single VRPTW route

```
def check_feasibility(route, customers, depot, Q):
    """Check capacity and time window feasibility of a route."""
    load = sum(customers[i]['demand'] for i in route)
    if load > Q:
        return False, None
    times = []
    t = depot['tw_open']
    prev = (depot['x'], depot['y'])
    for idx, c in enumerate(route):
        t += euclidean(prev, (customers[c]['x'], customers[c]['y']))
        if t > customers[c]['tw_close']:
            return False, None
        t = max(t, customers[c]['tw_open'])
        times.append(t)
        t += customers[c]['service_time']
        prev = (customers[c]['x'], customers[c]['y'])
    return True, times

def two_opt(route, customers, depot, Q):
    """Improve a single route by 2-opt moves."""
    improved = True
    best_route = route[:]
    while improved:
        improved = False
        for i in range(1, len(best_route)-1):
```

5.5 — Extensions of the VRPTW I

The basic VRPTW has inspired a rich family of extensions that model real-world logistics problems more faithfully. This section surveys the most important ones.

5.5.1 Split Deliveries

In the standard VRPTW, each customer is visited by exactly one vehicle. In the **Split Delivery VRPTW (SDVRPTW)**, a customer with large demand can be served by *multiple* vehicles, each delivering a fraction of the total demand. This is useful when a customer's demand exceeds a single vehicle's capacity.

The SDVRPTW relaxes the integrality of the coverage constraints:

$$\sum_{k \in K} \sum_{j \in V} x_{ij}^k \geq 1 \quad \forall i \in V'$$

(at least one visit, but possibly more). The total delivery across all visits must sum to the demand d_i .

5.5 — Extensions of the VRPTW II

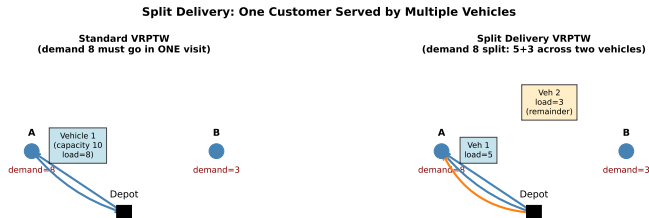


Figure: Split delivery example. In the standard VRPTW (left), customer A's demand of 8 must fit in one vehicle (capacity 10 — just feasible). In the SDVRPTW (right), a second vehicle can deliver the remaining demand, which can reduce total route length when the demand makes single-vehicle routing awkward.

5.5 — Extensions of the VRPTW III

5.5.2 Time-Dependent Travel Times

In urban logistics, travel times are not constant: they depend on the time of day due to traffic congestion. The **Time-Dependent VRPTW (TDVRPTW)** replaces the constant travel time t_{ij} with a function $t_{ij}(\tau)$ (t sub ij of tau), where τ (tau, Greek letter) is the departure time from node i .

The resource extension function for the time resource becomes:

$$T_j = \max(T_i + s_i + t_{ij}(T_i + s_i), a_j)$$

Notice that t_{ij} now depends on the departure time $T_i + s_i$. A common model piecewise-linearly approximates $t_{ij}(\tau)$ using road speed profiles from historical traffic data (GPS traces, loop detectors).

For example, in a metropolitan area, the travel from zone A to zone B takes 10 minutes at 6 am but 35 minutes during rush hour (8–9 am). A route that serves customer i at 7:50 am and needs to reach customer j in zone B should plan for 35 minutes, not 10.

The TDVRPTW makes the SPPRC pricing sub-problem harder but remains tractable with time-discretised arc travel times.

5.5 — Extensions of the VRPTW IV

5.5.3 Multi-Depot VRPTW (MDVRPTW)

Instead of a single depot, the **Multi-Depot VRPTW** has D depots. Each vehicle is assigned to one depot and must both start and end its route at that depot. Customers can potentially be served from any depot.

The multi-depot structure complicates both the assignment of customers to depots and the routing. A natural formulation replicates the single-depot model with D origin-destination depot pairs and adds a constraint that each vehicle returns to its home depot. The set-partitioning formulation extends cleanly: each feasible route is now associated with a specific depot, and the column generation sub-problem (SPPRC) runs separately for each depot.

5.5 — Extensions of the VRPTW V

5.5.4 Heterogeneous Fleet VRPTW

In practice, companies operate fleets with vehicles of different capacities, costs, and speeds. The **Heterogeneous Fleet VRPTW (HFVRPTW)** introduces vehicle types. Each vehicle type v has its own capacity Q_v , fixed cost f_v , and per-unit travel cost c_{ij}^v .

The three-index formulation extends by indexing binary variables as x_{ij}^k where vehicle k belongs to a specific type. The column generation approach separates sub-problems by vehicle type.

5.5 — Extensions of the VRPTW VI

5.5.5 Stochastic VRPTW

Real logistics operations face uncertainty: demands may be known only in distribution, travel times may vary, customers may cancel. The **Stochastic VRPTW** models such uncertainty.

Stochastic demands: Each customer i has a demand $\tilde{d}_i$ (tilde d sub i) that is a random variable known only in distribution. A *two-stage stochastic programming* model plans routes before demands are realised (first stage) and allows *recourse actions* after realisation (second stage): if a vehicle's load is exceeded at some customer, it returns to the depot to reload and resumes the route (at additional cost).

The objective minimises expected total cost:

$$\min C_{\text{routing}} + E[\text{recourse cost}]$$

where $E[\cdot]$ (E of expectation, the expected value) denotes the expectation over all demand realisations.

Stochastic travel times: Similarly, $\tilde{t}_{ij}$ is random (e.g., Gaussian with mean μ_{ij} and variance σ_{ij}^2). The solution must be robust: time windows are met with high probability, not just in expectation.

5.5 — Extensions of the VRPTW VII

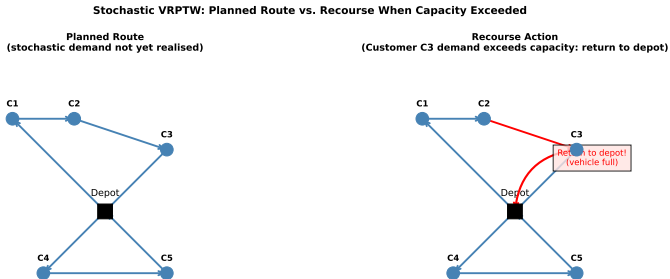


Figure: Stochastic VRPTW. The planned route (left) visits customers in a fixed sequence. When customer C3's actual demand exceeds the vehicle's remaining capacity, the vehicle returns to the depot (right, red arrow) to reload before continuing—a recourse action with additional travel cost.

5.5 — Extensions of the VRPTW VIII

5.5.6 Driver Considerations and Regulations

In practice, drivers are subject to legal regulations on driving time and rest periods (e.g., EU Regulation 561/2006 limits to 9 hours of driving per day). This adds another resource constraint to the SPPRC:

- **Continuous driving time:** Must not exceed a threshold (e.g., 4.5 hours) before a mandatory break of at least 45 minutes.
- **Total working time:** Maximum shift length per day.

These constraints make the pricing sub-problem harder (more resources) but remain tractable with multi-dimensional labelling. Reduced cost fixing becomes more powerful with more resources (tighter feasibility conditions prune more labels).

5.6 — Conclusions and Future Research I

Summary of the chapter. The VRPTW is a rich and practically important optimisation problem that has been the focus of intense research for over three decades. The main messages are:

- 1 **Exact methods.** Branch-and-Price, powered by column generation and the SPPRC, is the state-of-the-art exact approach. It can solve all 56 Solomon benchmark instances (100 customers) to optimality. Key advances include bidirectional labelling, dual stabilisation, and valid inequalities (subset-row cuts).
- 2 **Heuristics.** For large instances (200+ customers), metaheuristics dominate. The best-performing methods are hybrid approaches combining large neighbourhood search (ALNS) with population-based diversity mechanisms and efficient local search. Differences between top methods are small (within 1%), so the gap between heuristic and optimal is well understood.
- 3 **Benchmarking.** The Solomon (100-customer) and Gehring-Homberger (200–1000 customer) benchmark sets provide universally accepted comparison standards. Best-known solutions are maintained in public repositories.
- 4 **Extensions.** The VRPTW framework naturally accommodates split deliveries, multiple depots, heterogeneous fleets, time-dependent travel, stochastic demands, and driver regulations. Each extension requires adaptations to both the formulation and the algorithmic components.

5.6 — Conclusions and Future Research II

Future research directions.

- **Larger exact instances:** Current B&P methods struggle beyond 100 customers. Incorporating machine learning to guide branching decisions (e.g., learning variable selection from previous instances) is an active frontier.
- **Rich extensions:** Combining multiple extensions simultaneously (e.g., stochastic + multi-depot + time-dependent) remains largely unsolved exactly. Approximate dynamic programming and robust optimisation approaches are promising.
- **Real-time and dynamic VRP:** Logistics operations are increasingly dynamic: orders arrive throughout the day, traffic conditions change in real time. Online algorithms and re-optimisation heuristics are needed.
- **Green VRP with time windows:** Minimising fuel consumption or emissions rather than distance adds a non-linear objective. Electric vehicle routing with charging stops is a particularly active area.
- **Machine learning for heuristics:** Deep reinforcement learning (attention-based models, pointer networks) has shown promise on small VRP instances but scaling to VRPTW with 500+ customers remains an open challenge.

5.6 — Conclusions and Future Research III

- **Parallelism and GPU acceleration:** Large neighbourhood search and population-based methods are embarrassingly parallelisable. GPU-based implementations can evaluate millions of moves per second, potentially enabling exact-quality solutions on large instances.

Final Takeaway

The VRPTW demonstrates how a single, clean mathematical formulation—the set-partitioning model with column generation—unifies exact and heuristic methods: column generation solves the LP relaxation, branch-and-price finds integer solutions, and ALNS achieves near-optimal solutions heuristically. This unification of mathematical programming and metaheuristics is one of the great success stories of modern combinatorial optimisation.

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Chapter 5: The Vehicle Routing Problem with Time Windows

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Vehicle Routing: Problems, Methods, and Applications, 2nd ed.
Chapter 5, pp. 119–171.

Key topics covered:

- VRPTW formulations (three-index, two-index, set-partitioning)
- Column generation and Branch-and-Price (SPPRC, labelling)
- Construction heuristics (Solomon I1)
- Local search neighbourhoods (2-opt, Or-opt, cross-exchange)
- Metaheuristics (TS, SA, GLS, LNS/ALNS, ILS, EA)
- Extensions: split delivery, time-dependent, multi-depot, stochastic